

1. Introduction

The purpose of this study is to develop a starting point for discounting the value of multi-year player contracts. After attending a panel with the Minnesota Twins Baseball Operations department, I received one of the clearest non-proprietary answers I have heard on the subject. Their approach relies on two primary inputs: a player's contribution to winning and league-wide salary inflation.

This paper focuses on the second of those inputs. Specifically, it asks: what is league inflation (g), and what drives it?

League inflation refers to the rate at which the market value of player performance increases over time. It is a critical input in contract valuation, yet it is rarely defined explicitly and is often approximated using rough assumptions or general economic indicators. This study attempts to estimate league inflation directly and test several factors that may influence it.

The factors considered are:

- League revenue as a proxy for the overall economic size of MLB
- The structure of the Collective Bargaining Agreement (CBA)
- Broader U.S. economic conditions, including inflation (CPI) and unemployment

All regressions were conducted in R using annual data to evaluate how these factors relate to salary growth and, by extension, league inflation.

This paper is not intended to provide a definitive model, but rather to establish a framework for thinking about league inflation and to test whether commonly assumed drivers, such as revenue or macroeconomic conditions, actually explain it. Some additional analyses explore related questions, such as how insulated MLB revenues are from broader economic cycles. These are included as supporting context rather than central results.

The goal is to define league inflation in a way that is both empirically grounded and useful for valuation, while identifying the institutional and economic forces that cause it to change over time.

2. Valuation Framework and the Role of League Inflation (g)

"If you look at everything Warren Buffett and Charlie Munger say, if you take the word 'investing' out and put 'baseball' in, every sentence still works."

- Billy Beane, Former A's GM at 2022 Tipalti Illuminate tech conference

By this thought I am going to approach this as a value investor might, by combining what the Twins Discussed to the Discounted Cash Flows essential to determine value of a stock. Teams are comparing the present value of future on-field production to the cost of acquiring that production. In finance, this involves projecting future value and discounting it back to today using an appropriate rate.

In baseball, however, the problem has an important twist. The value of a player's performance is not fixed over time. It grows as the market price of performance increases. A player who produces the same level of performance in future years will typically be paid more, not because the performance improved, but because the market has changed.

This introduces a key parameter, league inflation (g), which represents the rate at which the dollar value of player production increases each year.

A simplified version of the valuation framework can be written as:

$$\text{Value} = \sum_{t=1}^T \frac{\text{Player Value}_t \times P_0 \times (1 + g)^t}{(1+r)^t}$$

where:

- Player Value_t is projected performance in year t under any chosen metric
- P_0 is the current market price of that performance
- g is league inflation
- r is the discount rate
- t is the number of years into the future, and T is the total contract length

In plain terms, this means that future production becomes more valuable in dollar terms as the market for player performance grows.

Rewriting the equation highlights the role of league inflation:

$$\text{Value} = \sum_{t=1}^T \frac{\text{Player Value}_t \times P_0}{(1 + r - g)^t}$$

This shows that teams are effectively working with a net discount rate. Growth in league pricing offsets part of the time discount, so future value is not discounted as heavily as it would be in a standard financial model.

The focus of this study is estimating g . While discount rates can be approximated using financial benchmarks, league inflation is specific to baseball. It reflects how the market for player performance evolves over time and is shaped by league revenues, institutional rules such as the CBA, and broader market behavior.

Understanding league inflation is therefore a necessary step before any meaningful discussion of discounting player contracts can take place. The remainder of this paper focuses on estimating g and testing the factors that influence it.

3. Data and Methodology

To estimate league inflation and test its potential drivers, this study constructs a panel of annual MLB financial and institutional variables spanning multiple decades. The goal is not to build a fully structural model, but to evaluate how salary growth relates to observable economic and institutional factors.

Data

The dataset combines four primary components:

- **Player Compensation**

League-wide average salary and payroll data were sourced from The Baseball Cube, which provides consistent historical coverage beginning in the late 1980s. Average salary is used as the primary proxy for the market price of player performance.

- **League Revenue**

Annual MLB revenue from 2001 onward was compiled from Forbes' team valuation reports, which are widely used in sports economics. Revenue prior to 2001 was reconstructed using reported media contracts, attendance trends, and known benchmark values. While these earlier estimates are less precise, they provide continuity for long-run analysis.

- **Collective Bargaining Agreements (CBA)**

CBA timing was incorporated as both a binary indicator and a regime variable. The binary variable captures the first full season under a new agreement, while the regime variable tracks the full period governed by each CBA. This distinction allows for testing both immediate and structural effects of labor agreements.

- **Macroeconomic Variables**

U.S. inflation (CPI) and unemployment were included as proxies for broader economic conditions. CPI data were sourced from the Bureau of Labor Statistics, and unemployment data were obtained from FRED as annual averages.

Measuring League Inflation

League inflation (g) is defined as the growth rate of average MLB salaries. Two approaches are used:

1. **Year-over-year growth**

Log differences in average salary provide annual growth rates:

$$g_t = \ln \left(\frac{\text{Salary}_t}{\text{Salary}_{t-1}} \right)$$

2. **Multi-year growth (CAGR)**

To estimate longer-term trends, compound annual growth rates are calculated over different horizons:

$$g = \frac{\ln (\text{Salary}_t / \text{Salary}_0)}{T}$$

This approach ensures consistency with compounding and reduces sensitivity to short-term volatility.

Outlier years associated with structural disruptions, such as 1990 and 2020, are excluded from short-term estimates to avoid distorting results.

Empirical Strategy

The analysis proceeds in two steps.

First, league salary levels are regressed on revenue, inflation, and CBA indicators to establish baseline relationships:

$$\log(\text{Salary}) = \beta_1 \cdot \text{Revenue} + \beta_2 \cdot \text{CBA} + \beta_3 \cdot \text{Inflation} + \varepsilon$$

Second, the model is extended to account for institutional structure by interacting revenue with CBA regimes:

$$\log(\text{Salary}) = \beta_1 \cdot \text{Revenue} + \beta_2 \cdot \text{CBA Regime} + \beta_3 \cdot (\text{Revenue} \times \text{CBA Regime}) + \varepsilon$$

This specification tests whether the relationship between revenue and salaries changes under different labor agreements, rather than assuming a constant effect over time.

Additional regressions examine whether MLB revenues themselves are sensitive to macroeconomic conditions, providing context for how externally driven league inflation might be.

Limitations

Several limitations should be noted. Historical revenue data 2024-2001 are obtained through Forbes annual article naming franchise valuations and their underlying performance. This is as good as it gets outside of public information, like that from BATRA. However, prior to 2001 revenues are reconstructed from the various elements previously stated, they are a best estimate. The CBA variables assume that economic effects occur at the season level, while they may phase in gradually. The analysis also relies on league-wide aggregates, which may mask variation across teams and player segments.

Despite these limitations, the dataset provides a consistent and transparent framework for examining long-run relationships between salaries, revenues, and institutional structure in MLB.

4. Drivers of League Inflation: Empirical Results

The first step in understanding league inflation is identifying what drives salary growth. A natural starting point is to test whether salaries respond primarily to economic conditions, such as league revenue or general inflation.

Baseline Results

A baseline regression of average MLB salaries on league revenue, inflation, and a CBA indicator suggests that salary growth is largely explained by economic factors. In this specification, league revenue enters with a positive and highly significant coefficient

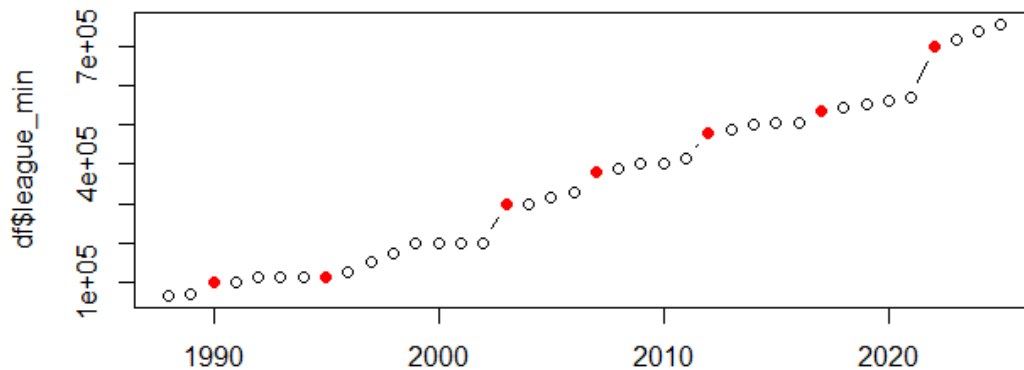
(1.37×10^{-10} , $p < 0.001$), while the CBA indicator is small and statistically insignificant (-0.062 , $p = 0.565$). Inflation enters with a negative and statistically significant coefficient (-0.063 , $p = 0.032$). The model explains a large share of variation in salaries ($R^2 = 0.83$).

Taken at face value, these results support a simple conclusion: salary growth appears to follow league revenue, with little independent role for institutional factors such as the CBA. This is the familiar “it’s the economy” explanation.

However, this interpretation assumes that the effect of revenue is constant over time and independent of the institutional environment. That assumption is unlikely to hold in a system where labor agreements explicitly govern how revenue is distributed.

Evidence of Institutional Structure

Examining salary structure directly suggests a different story. Minimum salary data shows that compensation does not evolve smoothly over time, but instead increases in discrete steps aligned with new CBA periods. These step changes reflect renegotiated minimum salaries, arbitration rules, and other institutional features that persist for multiple seasons.



This pattern indicates that CBAs shape the structure of compensation over time, rather than generating one-time effects in the year they are signed. A simple indicator for “new CBA year” therefore fails to capture the true role of labor agreements.

Interaction Model: Revenue Through the CBA

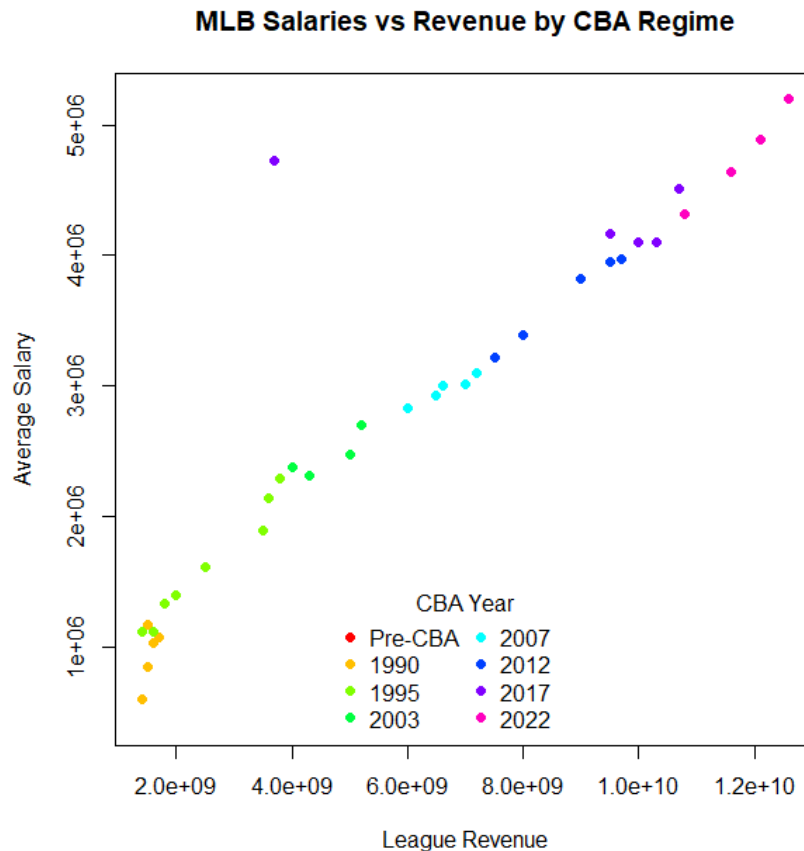
To account for this structure, the model was re-specified to allow the relationship between revenue and salaries to vary across CBA regimes:

$$\log(\text{Salary}) = \beta_1 \cdot \text{Revenue} + \beta_2 \cdot \text{CBA Regime} + \beta_3 \cdot (\text{Revenue} \times \text{CBA Regime}) + \varepsilon$$

This specification yields a markedly different result. The interaction between revenue and CBA regime is positive and highly significant (2.65×10^{-11} , $p < 0.001$), while the standalone

effect of revenue becomes statistically insignificant (-2.32×10^{-11} , $p = 0.213$). The CBA regime variable itself is large and highly significant (-0.325 , $p < 0.001$), and model fit improves substantially ($R^2 = 0.96$).

These results indicate that the effect of revenue on salaries is not constant, but depends on the governing CBA. In practical terms, newer CBAs increase the sensitivity of salaries to revenue, allowing a greater share of league income to flow to players.



Interpretation

This resolves the apparent contradiction in the baseline results. The strong correlation between revenue and salaries does not imply that revenue alone determines compensation. Rather, it reflects the fact that CBAs are structured around league revenues.

When the institutional framework is omitted, regression attributes salary growth to revenue. When it is included, the mechanism becomes clear: the CBA determines how revenue is translated into player salaries.

From a baseball perspective, this reflects the evolution of the labor market. Earlier CBAs imposed tighter constraints on compensation, while more recent agreements expanded arbitration systems, increased minimum salaries, and adjusted competitive balance tax thresholds. These changes have made player compensation more responsive to league financial performance.

Implications for League Inflation

These findings suggest that league inflation is not a purely economic variable. Instead, it is the result of an interaction between revenue growth and institutional structure.

As a result, league inflation should be expected to vary across time depending on the prevailing CBA and market environment, rather than following a single stable trend.

5. Estimating League Inflation (g)

Having established that league inflation is shaped by both revenue growth and institutional structure, the next step is to estimate its magnitude.

League inflation (g) is defined as the growth rate of average MLB salaries. To ensure consistency with compounding, growth rates are estimated using a compound annual growth rate (CAGR) framework rather than simple averages of year-over-year changes. Specifically:

$$g = \frac{\ln (\text{Salary}_t / \text{Salary}_0)}{T}$$

where T is the number of years in the sample.

This approach provides a long-run, compounding-consistent measure of salary growth. However, because league inflation varies over time, estimates are calculated across multiple horizons to capture both historical and recent conditions.

Estimated Growth Rates

The resulting estimates are:

Period	Estimated g
3-Year	5.65%
5-Year	5.52%
10-Year	3.01%
Full Sample	6.68%

These results show that league inflation is highly sensitive to the chosen time horizon.

The full-sample estimate of 6.68% reflects long-run growth, but is heavily influenced by periods of rapid expansion in the late 1980s and 1990s. In contrast, the 10-year estimate of 3.01% provides a more stable view of the modern labor market under recent CBA conditions. Shorter-term estimates over the past 3 to 5 years are higher, suggesting a recent acceleration in salary growth following a period of stagnation.

Interpretation

The variation across time horizons highlights an important feature of league inflation: it is not constant. Instead, it reflects the interaction of revenue growth, labor agreements, and market structure.

From a practical perspective, this means that a single historical average is unlikely to be an appropriate input for valuation models. A long-run estimate may overstate expected growth in the current environment, while a short-run estimate may overreact to recent market conditions.

Forward-Looking Scenarios

Given this variation, it is more appropriate to treat league inflation as a range of plausible outcomes:

- **Bear Case: ~3.0%**

Based on the 10-year estimate, this reflects a constrained labor market under current CBA rules, including competitive balance tax effects and reduced spending in the mid-tier free agent market.

- **Base Case: ~4.0–4.5%**

A blended estimate that assumes moderate revenue growth and partial transmission of that growth into salaries.

- **Bull Case: ~5.5–6.0%**

Based on recent 3–5 year trends, this scenario assumes stronger revenue growth and a more player-favorable distribution of income in future CBAs.

Summary

League inflation in MLB is best understood as a time-varying parameter rather than a fixed constant. While long-run growth averages approximately 6.7% per year, more recent data suggests a lower steady-state closer to 3–5%, with meaningful variation depending on economic conditions and institutional structure.

For valuation purposes, this supports the use of scenario-based assumptions for g , rather than reliance on a single historical estimate.

6. Implications for Player Valuation and Discounting

The results have direct implications for how teams approach the valuation of multi-year player contracts.

Front offices typically project a player's future production and assign a dollar value to that production based on current market prices. That value is then adjusted for expected changes in the market and discounted back to present value. In practice, this means that two inputs are doing most of the work: projected performance and league inflation.

The findings in this study suggest that league inflation should not be treated as a fixed or externally given parameter. Instead, it is a function of both economic conditions within the league and the institutional structure that governs how those conditions translate into player compensation.

A key takeaway is that league inflation is not well approximated by general economic indicators such as CPI. Earlier results show that inflation is either weakly related or negatively related to salary growth, indicating that baseball operates under a distinct economic system. Using general inflation as a proxy for salary growth would therefore misstate the expected value of future contracts.

At the same time, while league revenues are strongly correlated with salaries, that relationship is mediated by the CBA. The interaction results demonstrate that the sensitivity of salaries to revenue varies across labor regimes. This implies that forward-

looking estimates of league inflation must account for the current institutional environment, rather than assuming a stable historical relationship.

From a practical standpoint, this reinforces the need to use a range of assumptions for g , rather than a single point estimate. A lower growth assumption, such as the 10-year estimate of approximately 3%, reflects a more constrained labor environment and results in heavier discounting of future value. A higher assumption, closer to recent 3–5 year trends, implies a more favorable market for players and increases the present value of long-term contracts.

In the valuation framework introduced earlier, this directly affects the effective discount rate. Because future player value grows with league inflation, higher values of g reduce the net discount rate and increase the present value of future performance. Conversely, lower values of g increase the effective discounting of future years.

The central implication is that small changes in league inflation assumptions can have large effects on contract valuation, particularly for long-term deals. As a result, accurately estimating and contextualizing g is as important as projecting player performance itself.

More broadly, these findings suggest that player valuation is not purely a financial exercise, but one that is deeply tied to the institutional structure of the league. Understanding how revenue, labor agreements, and market behavior interact to shape league inflation is therefore essential for making informed decisions about long-term contracts.

An Annex of Thoughts and Detours

A. Supplementary Analysis: MLB Revenues and the Macroeconomy

As a complement to the primary analysis, this study examines whether MLB revenues are sensitive to broader macroeconomic conditions. This question is relevant because, if league revenues are strongly tied to the business cycle, then league inflation could be indirectly influenced by external economic factors.

To test this, league revenue was regressed on U.S. inflation (CPI) and unemployment rates.

Results

The results show weak explanatory power ($R^2 = 0.13$), indicating that macroeconomic variables account for little of the variation in MLB revenues. While both inflation (-0.157 , $p = 0.061$) and unemployment (-0.109 , $p = 0.165$) enter with negative coefficients, consistent with theoretical expectations, neither variable is statistically significant at conventional levels.

When a time trend is introduced to account for the long-term growth of league revenues, both variables become economically negligible. This suggests that any apparent relationship in simpler models is driven primarily by shared time trends rather than a causal link.

Interpretation

These findings suggest that MLB revenues are relatively insulated from short-term macroeconomic fluctuations. While economic downturns may exert some pressure on attendance or local revenue streams, the overall effect on league-wide revenue is muted.

This insulation likely reflects the structure of MLB's revenue model, which is heavily driven by long-term media contracts, centralized revenue streams, and institutional arrangements. These features reduce sensitivity to short-term changes in consumer spending and employment conditions.

Implications for League Inflation

The limited relationship between macroeconomic variables and league revenue has an important implication for league inflation. If revenues themselves are not strongly tied to the broader economy, then league inflation is unlikely to be driven by general economic conditions such as CPI.

Instead, league inflation appears to be determined primarily within the baseball ecosystem, through the interaction of revenue growth and institutional rules governing how

that revenue is distributed. This reinforces the earlier conclusion that league inflation should be modeled as a league-specific variable, rather than approximated using general economic indicators.

Scope of This Analysis

This section is intended as a supplementary check rather than a core component of the study. The primary focus remains on estimating league inflation and identifying its direct drivers. However, the results here provide useful context by showing that those drivers operate largely independently of the broader macroeconomic environment.

B. Future Research and Open Questions

This study is intended as a starting point rather than a complete model of league inflation. Several areas emerged during the analysis that warrant further investigation.

Distribution of Spending Across Teams

One open question is how spending behavior at the top of the market influences league-wide salary growth. MLB does not operate as a perfectly competitive market. A small number of teams account for a disproportionate share of total payroll, particularly in recent years.

This raises the question of whether league inflation is driven primarily by aggregate revenue, or by the behavior of high-spending teams. A useful extension of this study would examine whether changes in the spending patterns of the top decile of teams lead or amplify broader market movements.

Framed differently, there may be a “price-setting” mechanism within the league, where aggressive spending by a small number of teams establishes benchmarks that the rest of the market responds to. Whether this dynamic resembles a competitive market, a coordinated equilibrium, or something closer to a form of informal monetary policy is an open question.

Measurement of League Value

League revenue was used in this study as a proxy for the overall economic size of MLB. While this is a practical and widely accepted measure, it may not fully capture the underlying value of the league.

Future work could explore alternative measures, such as franchise valuations, media rights growth, or revenue composition. In particular, understanding how different components of

revenue translate into player compensation could provide a more detailed view of how league inflation is generated.

Heterogeneity Within the Labor Market

This study uses league-wide average salary as a proxy for the price of player performance. However, the labor market is not uniform.

Recent trends suggest that:

- top-tier players continue to command increasing salaries
- mid-tier free agents have experienced stagnation or decline
- Bottom tier raise dramatically with the CBA but have also increased a decent amount in recent years. Is this all CBA?

A more granular analysis could examine whether league inflation differs across segments of the market. If growth is concentrated at the top, then a single estimate of g may obscure important variation that affects valuation differently depending on player type.

Dynamics of CBA Effects

The analysis treats CBAs as discrete regime changes, where new rules take effect at the beginning of a season. In reality, the effects of a CBA may evolve over time as teams and players adjust behavior.

Future work could explore:

- lagged effects of CBA changes
- differences between early and late years within a CBA period
- how expectations of future CBAs influence current behavior

This would provide a more dynamic view of how institutional rules shape salary growth.

Interaction Between Revenue Growth and Institutional Structure

One of the central findings of this study is that revenue affects salaries through the CBA rather than independently. A natural extension would be to model this relationship more explicitly.

Rather than treating revenue and CBA structure as separate variables, future work could attempt to estimate how specific provisions within CBAs alter the transmission of revenue into player compensation. This would move the analysis closer to a structural model of league inflation.

Summary

These extensions highlight that league inflation is not a simple or purely economic variable. It is shaped by market structure, institutional design, and strategic behavior across teams.

Understanding these dynamics more fully would improve both the estimation of g and its application in player valuation models.

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